
CISCO FTD

Application Visibility & Control FMC-Managed Configuration Guide

A lab walkthrough demonstrating Layer 7 Application Visibility & Control (AVC) on Cisco FTD — from understanding deep packet inspection and application-level granularity, through ACP rule creation, to live testing with Discord showing how AVC can block voice calls while allowing chat.

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Category: Network Security Lab · Portfolio Project

Platform: Cisco Firepower (FMC + FTD · AVC + SSL Decryption)

Prerequisite: SSL/TLS Decryption (configured in previous lab)

1. Introduction

Application Visibility & Control (AVC) in Cisco Secure Firewall Threat Defense (FTD) is a capability that lets the firewall identify, classify, monitor, and control applications at Layer 7 — not just by IP address or port number. This goes far beyond traditional firewall rules that rely on TCP/UDP port matching.

This lab builds on the SSL/TLS decryption configured in the previous lab. SSL decryption is a prerequisite for AVC because without it, the firewall cannot inspect encrypted HTTPS traffic to accurately identify the application inside.

► 1.1 What Does AVC Do?

Instead of trusting ports (e.g., TCP/443 = HTTPS), FTD uses Deep Packet Inspection (DPI) and Snort application detectors to identify the actual application running inside the traffic flow. This enables security policies like:

- Allow Microsoft Teams but block remote administration tools
- Allow HTTP browsing but block file uploads
- Allow Discord chat but block Discord voice calls
- Block specific application sub-features without blocking the entire application

► 1.2 Traditional Firewall vs. AVC

Aspect	Traditional Firewall (L3/L4)	AVC-Enabled Firewall (L7)
Identification	IP address + port number	Application signature via DPI
Granularity	Allow/block entire port	Allow/block specific app features
Evasion resistance	Apps can use port 443 to bypass	Identifies app regardless of port
Example	Allow TCP/443	Allow Teams, block Discord Call
Visibility	Source/destination/port	App name, category, bandwidth, user

► 1.3 Application Control Granularity

AVC supports sub-feature-level control for applications that expose multiple functions. This is the key differentiator — you can selectively allow or block specific capabilities:

- Allow Facebook chat but block Facebook file upload
- Allow HTTP GET requests but block HTTP POST
- Allow Discord messaging but block Discord Call (voice/video)

This granularity is what makes AVC essential for enterprise security policies where blanket blocking is too restrictive.

► 1.4 Visibility & Reporting

AVC provides rich visibility through FMC dashboards and reports:

- Top applications by bandwidth consumption
- Application usage broken down by user identity
- Trend and timeline analysis for traffic patterns
- All visible in FMC → Analysis → Application Statistics

2. Lab Environment

This lab uses the same topology as the previous SSL Decryption guide. The FTD is registered to FMC with interfaces, ACP, NAT, static routing, and SSL decryption already configured.

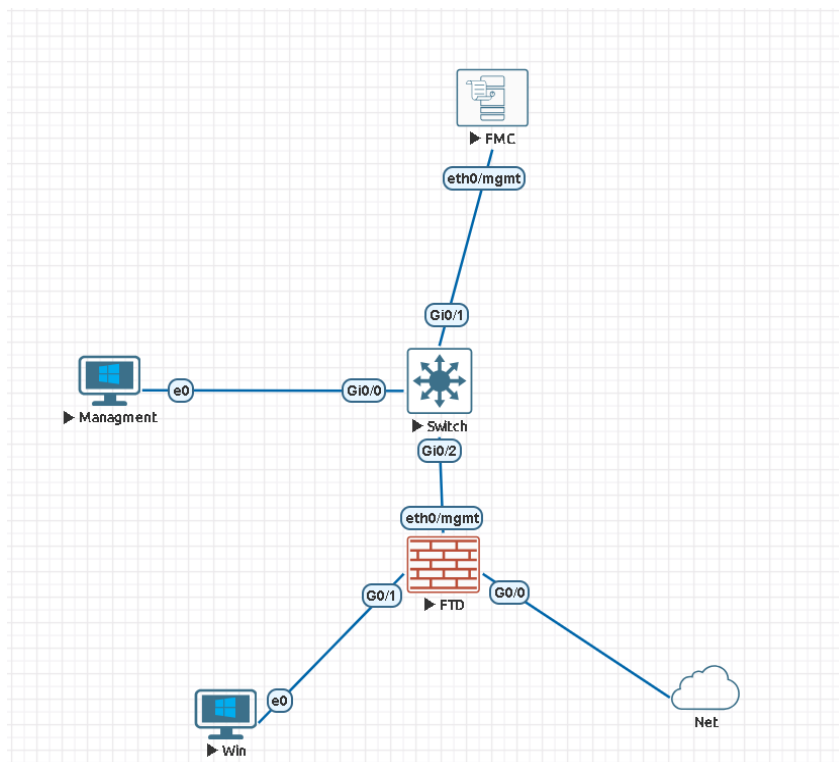


Figure 1 — Lab Topology (same as SSL Decryption lab)

Discord is used as the test application to validate AVC functionality — both its website (HTTPS) and its desktop voice calling feature.

3. AVC Configuration

► 3.1 Create the Application Block Rule

To block Discord (both the website and the voice call feature), create an ACP rule targeting the Discord application signatures:

1. Navigate to Policies → Access Control → Access Control Policies.
2. Edit your main policy (e.g., to_internet).
3. Click Add Rule.

4. Configure the rule:
 - **Name:** New-Rule-#2-BLOCK
 - **Action:** Block
 - **Source Zone:** Inside_zone
 - **Destination Zone:** Outside_zone
5. Go to the Applications tab.
6. Search for "discord" — select both Discord and Discord Call.
7. Save the rule.

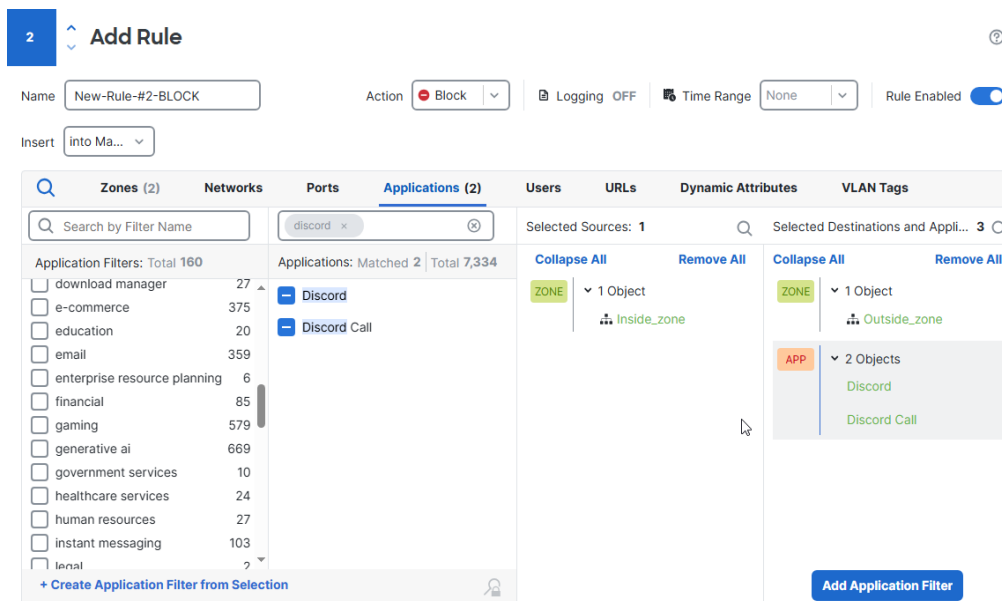


Figure 2 — AVC rule: Block Discord + Discord Call (Inside → Outside)

3.2 Rule Order

Ensure the Block rule is positioned above any Allow rule in the ACP. Rules are evaluated top-down — the first matching rule wins. The Discord block rule must be processed before any general Allow rule that would otherwise permit the traffic.

3.3 Deploy

Click Deploy → Deploy to FTD. Wait 1–3 minutes for the policy to propagate to the appliance.

4. Testing & Verification

► 4.1 Test 1 — Discord Website

From the test client, attempt to browse to <https://discord.com/download>. The connection should be reset (ERR_CONNECTION_RESET), confirming the AVC rule is blocking the Discord application:

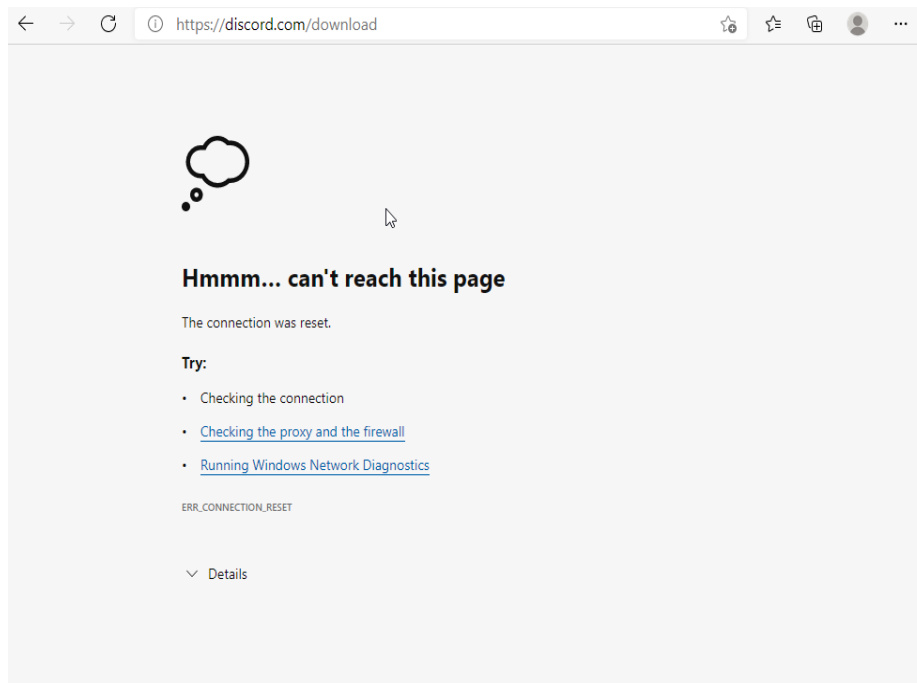


Figure 3 — Discord website blocked: ERR_CONNECTION_RESET

► 4.2 Test 2 — Discord Voice Call (Granularity Demo)

Now, modify the ACP rule to allow the Discord application but keep blocking Discord Call. This demonstrates the core value of AVC — sub-feature-level control.

After redeploying with only Discord Call blocked:

- Discord chat and messaging → allowed (working normally)
- Discord voice/video call → blocked (shows "No Route" error)

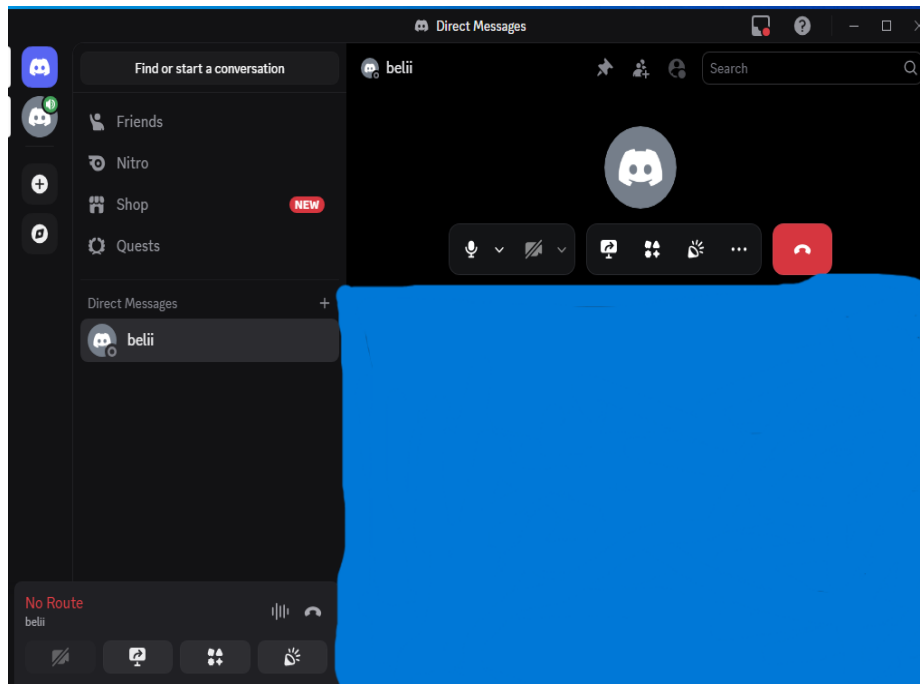


Figure 4 — Discord Call blocked ("No Route") while chat remains functional

This is the main point of AVC: the ability to block specific application sub-features (voice calls) while allowing the rest of the application (chat/messaging) to function normally.

5. AVC Use Cases

AVC is particularly valuable in enterprise and OT/ICS environments where granular application control is required:

► 5.1 Enterprise IT

- Allow Microsoft 365 suite but block personal cloud storage (Dropbox, Google Drive)
- Allow Zoom/Teams for business meetings but block gaming applications
- Monitor shadow IT — identify unauthorized applications on the network

► 5.2 OT/ICS Security

- Allow specific SCADA/HMI applications while blocking general internet browsing
- Block remote desktop tools (TeamViewer, AnyDesk) on OT network segments
- Monitor and control file transfer applications to prevent data exfiltration

► 5.3 Compliance & Policy Enforcement

- Block peer-to-peer applications (BitTorrent, uTorrent) across the organization
- Restrict social media to specific user groups or time windows
- Generate compliance reports showing application usage by user and department

6. AVC + SSL Decryption Relationship

AVC and SSL Decryption work together to provide complete application visibility:

Scenario	AVC Accuracy	Why
Without SSL Decryption	Limited — identifies app by SNI/header only	Cannot see inside encrypted payload
With SSL Decryption	Full — identifies app by payload inspection	FTD sees decrypted content via DPI
Sub-feature control (e.g., Discord Call)	Requires SSL Decryption	Voice/video signaling is inside TLS

For accurate Layer 7 application identification and sub-feature control, SSL Decryption is essential. Without it, AVC falls back to SNI-based identification which cannot distinguish between application sub-features.

7. Summary

Component	Details
Feature	Application Visibility & Control (AVC)
Layer	Layer 7 — Deep Packet Inspection (DPI)
Prerequisite	SSL/TLS Decryption for full accuracy
Detection Engine	Snort application detectors
Granularity	Sub-feature level (e.g., allow chat, block calls)
Test Application	Discord (website + voice call)
Key Result	Discord Call blocked while chat remains functional
Managed By	FMC → Access Control Policy → Applications tab
Visibility	FMC dashboards: top apps, bandwidth, user-level reporting
Typical Use Cases	Shadow IT detection, OT app control, compliance enforcement